BRAINSTORM GROUP (BSG) MAT101 MOCK TEST I

SET THEORY

1. Which of the following set description is not correct?
   A. $B = set\ of\ geometry\ book$
   B. $A = \{x; even\}$
   C. $H = \{1, 2, 3: 4\}$
   D. $S = \{a, b, c\}$

2. What is the value of $n(\{a, a, b\})$
   A. 0 B. 1
   C. 2 D. 3

3. List the element of the set $X = \{n^2 + 1;\ n = 0, 1, 2, 3\}$
   A. $\{0, 2, 4, 9\}$
   B. $\{1, 2, 5, 10\}$
   C. $\{0, 1, 2, 3\}$
   D. $\{0, 2, 5, 10\}$

4. If $S = \{2, 4, 6\}$ and $T = \{1, 2, 3, 4\}$, find $S \cup T$
   A. $\{1, 2, 3, 4, 5\}$
   B. $\{1, 2, 3, 4, 5, 6\}$
   C. $\{1, 2, 3, 4, 6\}$
   D. $\{2, 3, 4, 6\}$

5. Given the universal set $U = \{1,\ 2,\ 3,\ 4,\ 5,\ 6,\ 7,\ 8\}$ where $A = \{1,\ 3,\ 5,\ 7\}, B = \{1,\ 2,\ 5,\ 7\}$ and $C = \{3,\ 6,\ 7,\ 8\}$. List the members of $A^c \cap B \cap C^c$
   A. $\{1\}$ B. $\{2\}$ C. $\{3\}$
   D. $\{4\}$ E. $\{5\}$

6. If $F = \{x: (2 < x \leq 5)\ \cup\ (9 < x < 12)\}$, list the elements of set F
   A. $\{2, 3, 4, 5, 9, 10, 11, 12\}$
   B. $\{3, 4, 5, 9, 10, 11\}$
   C. $\{3, 4, 5, 10, 11\}$
   D. $\{3, 4, 5, 10, 11, 12\}$

7. If $X = \{2, 4, 6\}$, determine power set
   A. 8 B. 6
   C. 4 D. 3

8. Find the cardinality of set $X = \{I, N, T, E, R, C, O, N, T, I, N, E, N, T, A, L\}$
   A. 9 B. 10
   C. 15 D. 16

9. Find the cardinality of $n\ \epsilon\ \mathbb{N} : -5 < 2n < 12$
   A. 8 B. 9
   C. 12 D. 16

10. In a village of 650 youths, 250 are girls, 425 of the youths like music. If 90 girls do not like music, how many boys do like music?
    A. 135 B. 160
    C. 175 D. 265
    E. 335

11. $P - Q'$ is same as
    A. $(P \cap Q)$
    B. $(P \cap Q)'$
    C. $(P \cap Q')$
    D. $(P' \cap Q)$

12. If $X = \{prime\ factors\ of\ 84\}$ and $Y = \{prime\ \ factors\ of\ 315\}$, the elements of $X \cup Y$ and $X \cap Y$ are respectively.
    A. $\{3, 4, 5, 7, 9\}$ and $\{3, 7\}$
    B. $\{3, 4, 5, 7\}$ and $\{5, 7\}$
    C. $\{2, 3, 5, 7, \}$ and $\{3, 7\}$
    D. $\{2, 3, 5, 7, 9\}$ and $\{2, 7\}$

13. If $R = \{x: x^2 = 16, x > 5\}$, the R is equal to
    A. 0 B. $\{0\}$
    C. $\{\emptyset\}$ D. $\emptyset$

14. If $\mathcal{U} = \{integers \leq 20\}, X = \{multiples\ of\ 4\}, X = \{multiples\ of\ 3\}$. The element of $X' \cap Y$ are
    A. $\{12\}$
    B. $\{3, 6, 9, 12, 15, 18\}$
    C. $3, 6, 9, 15, 18\}$
    D. $4, 8, 16, 20\}$

MAPPING AND FUNCTION

15. If $S = \{2, 4, 6\}$ and $T = \blacksquare$, find $S \cap T$
    A. $\{2, 4\}$ B. $\{4, 6\}$
    C. $\{1, 2\}$ D. $\{2, 6\}$

16. 33 students in a class play either football or basketball. If 22 students play football and 18 play basketball, how many students play football only?
    A. 7 B. 11
    C. 15 D. 17

17. In a class of 60 students, 40 speak Hausa while 19 speak Igbo and 7 students speak both languages. How many students speak at least one language?
    A. 8 B. 42
    C. 45 D. 52

18. Which of the following is an empty set?
    A. $\{x: \ is\ a\ real\ number\ and\ x^2 - 1 = 0\}$
    B. $\{x: \ is\ a\ real\ number\ and\ x^2 + 1 = 0\}$
    C. $\{x: \ is\ a\ real\ number\ and\ x^2 - 9 = 0\}$
    D. $\{x: \ is\ a\ real\ number\ and\ x^2 = x + 2\}$

19. If $P \supset Q$, then $(P \cup Q) \cap P$ is
    A. $\emptyset$ B. P
    C. Q D. $\mathcal{U}$

20. In a market place all the people buy or sell things or do both. If 70% of the people do buy and 90% do sell, what percentage of people do both?
    A. 60% B. 20%
    C. 50% D. 30%

21. If $f: x = 2x - 3$ and $g: x = 3x + p$, for what value of constant p is $fog = gof$.
    A. -3 B. 3
    C. -2 D. -6

22. The function f is given as $f(x) = ln(4x - 2)$, find in $f^{-1}(x)$ its simplest form
    A. $\frac{1}{4}(e^x - 2)$ B. $-\frac{1}{4}(e^x - 2)$
    C. $\frac{1}{4}(e^x + 2)$ D. $-\frac{1}{4}(e^x + 2)$

23. If $f: x = 1 + 2x$ and $g: x = \frac{1}{1+x}$, find $f^{-1}g^{-1}\left(\frac{2}{3}\right)$
    A. $\frac{1}{4}$ B. $-\frac{1}{2}$
    C. $-\frac{1}{4}$ D. $-\frac{3}{4}$

24. Given $f(x) = 2x^2$ and $g(x) = 3x + 1$, Compute $(fog)(2)$
    A. 25 B. 49
    C. 98 D. 89

25. If $f: x \to \frac{2}{3(2-7x)}, x \neq \frac{2}{7}$, find $f^{-1}(x)$
    A. $\frac{6x-2}{21x}$ B. $\frac{2-6x}{21x}$
    C. $\frac{6x-2}{7x}$ D. $\frac{2-6x}{7x}$

26. Two functions f and g are defined on the set R of real numbers by: $f(x) = x^2 - 2x - 1$ and $g(x) = x - 1$. Find the value of x for which $f(x) = g(x) - 2$
    A. 0 *or* 1 B. 1 *or* 2
    C. 0 *or* 2 D. 1 *or* 3

27. Two functions f and g are defined on the set R of real numbers as follows: $f: x = 2x - 3$ and $g: x = x^2 + 5$. Find $f[g(x - 1)]$
    A. $2x^2 - 4x + 12$ B. $2x^2 - 4x + 9$
    C. $x^2 - 2x + 6$ D. $2x^2 - 2x + 12$

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28. Given that image of p under the mapping $x \rightarrow 3x + 2$ is -10, what is the value of p?
A. -4 B. -3
C. -2 D. -1

29. If every element in the domain has different image in the co-domain, such type of mapping is called
A. constant mapping
B. onto mapping
C. one-one mapping
D. composite mapping

30. A mapping f is called _____ if every element of the co-domain is an image of at least one element in the domain?
A. constant mapping
B. onto mapping
C. one-one mapping
D. composite mapping

31. The rule that assigns an element to two or non-empty set is
A. logic B. set
C. mapping D. rules

32. Given the inverse of a function to be $f(x) = \sqrt{4ax + b}$, find the main function
A. $\frac{x^2+b}{4a}$ B. $\left(\frac{x-b}{4a}\right)^2$
C. $\frac{x^2-b}{4a}$ D. $\left(\frac{x+b}{4a}\right)^2$

33. Find the inverse of $f: x = \frac{x+1}{x-2}$.
A. $\frac{2x+1}{x-1}$ B. $\frac{2x+1}{x+1}$
C. $\frac{2x-1}{x-1}$ D. $\frac{2x-1}{x+1}$

34. Determine the zero of the function defined by $h(x) = x^2 + x - 6$.
A. -2 B. 3
C. 1 D. 2

35. Given that $f(x) = x^5 + 4x^4 - 6x^2 + 2x + 2$, find $f(-1)$.
A. -3 B. -2
C. -1 D. 1

36. What is the set of values of x for which the mappings $x \rightarrow x^2$ and $x \rightarrow x^3$ have the same functional values?
A. $\{-1, 0, 1\}$ B. $\{1\}$
C. $\{0, 1\}$ D. $\{2\}$

37. The functions g and f are given by $g(x) = 4 - 3x$ and $f(x) = x^2 + ax + b$, where a and b are non-zero constants. Given that $fg(2) = -5$ and $gf(2) = -29$, find the value of a and b.
A. $a = -1, b = 4$
B. $a = 1, b = -4$
C. $a = 4, b = 1$
D. $a = 4, b = -1$

38. The function f is given as $f(x) = \frac{x+6}{x+2}, x \in \mathbb{R}, x \neq -2$, solve the equation $f(x) = f^{-1}(x)$
A. $-3, -2$ B. $3, -2$
C. $-3, 2$ D. $3, 2$

39. The function h is given as $h(x) = \frac{x}{x-1}, x \in \mathbb{R}, x \neq 1$, find in its simplest form the composition $hh(x)$
A. $\frac{x-1}{x}$ B. $x - 1$
C. $1 - x$ D. x

40. What is the name of the platform organizing this test?
A. Brainstorming Group
B. Brainstormers Group
C. Brainstorm Group
D. Brainstormer Group

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ANSWERS							
1.	C	11.	A	21.	D	31.	C
2.	C	12.	C	22.	C	32.	C
3.	B	13.	D	23.	C	33.	A
4.	C	14.	C	24.	C	34.	D
5.	B	15.	A	25.	A	35.	A
6.	C	16.	C	26.	B	36.	C
7.	A	17.	D	27.	B	37.	D
8.	A	18.	B	28.	A	38.	C
9.	A	19.	B	29.	C	39.	D
10.	D	20.	A	30.	B	40.	C

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